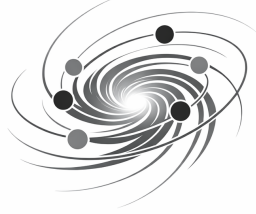


Part of the Monistic Continuum Model (MCM)



Electron Entanglement (EE)

Spectral and Topological Structures

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$$|\Psi_{12}\rangle = \hat{P}_{\text{QM}}(\mathcal{K}_{12}), \quad \mathcal{K}_{12} = \bigcup_{i=1}^N c_i, \quad c_i \subseteq W_n.$$

The quantum-mechanical two-electron configuration arises as a spectral projection of the underlying coupling geometry. The structure $\mathcal{K}_{12}$ is the union of all continuous sub-geometries c_i , each of which emerges from the vortex modes W_n .

Abstract

This work develops the continuum-geometric foundations of electron entanglement within the Monistic Continuum Model (MCM). In contrast to standard quantum mechanics, where entanglement is an algebraic property of tensor-product Hilbert spaces, the MCM interprets entanglement as a geometric and dynamical feature of a continuous substrate

$$C : M \times \Lambda_\Omega \rightarrow V,$$

organized by local and nonlocal operators. Nonlocal coherence generated by the operator D_{nl} produces non-separable correlation kernels

$$K(x, y; \Lambda_\Omega) \neq K(x; \Lambda_\Omega)K(y; \Lambda_\Omega),$$

leading to mixed spectral modes of the correlation operator $\mathcal{K}_{\Lambda_\Omega}$. This establishes a continuum-based mechanism for quantum-equivalent entanglement.

The main theorem demonstrates that nonlocal continuum coherence necessarily generates mixed eigenfunctions and non-separable kernel structures, providing a geometric origin for entanglement. The model predicts experimentally accessible deviations from standard QM, including correlation plateaus, geometry-dependent entanglement strength, non-monotonic coherence–distance curves, and persistent post-measurement correlations.

The MCM reproduces all empirically confirmed quantum phenomena while offering a unified geometric substrate linking quantum mechanics, quantum field theory, and general relativity. Electron entanglement emerges as a spectral projection of underlying coupling geometries, revealing entanglement as a structural property of the continuum rather than an abstract algebraic construct.

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1 Explanation: On the Title Equations

This section explains the meaning of the equations on the title page.

The two equations on the title page describe how the *entanglement of two electrons* arises from the geometry of the continuum.

1.1 1. The first equation

$$|\Psi_{12}\rangle = \hat{P}_{\text{QM}}(\mathcal{K}_{12})$$

This formula states: The joint quantum state of the two electrons, represented by $|\Psi_{12}\rangle$, originates from a deeper geometric structure $\mathcal{K}_{12}$. The operator $\hat{P}_{\text{QM}}$ means “projection into the quantum world.” It translates the geometric coupling of the electrons within the continuum into the state observed in quantum mechanics. In other words: Entanglement is not an abstract computational construct but the visible result of a real geometric connection.

1.2 2. The second equation

$$\mathcal{K}_{12} = \bigcup_{i=1}^N c_i, \quad c_i \subseteq W_n$$

This equation explains what the coupling structure $\mathcal{K}_{12}$ consists of. The individual c_i are small subregions or “vortex fragments” of the continuum. The symbol $\bigcup$ means “union” — all these subregions together form the entire coupling geometry. Each c_i lies within a vortex mode W_n , a geometric structure that represents the electron in the MCM.

1.3 3. Overall Idea

The two equations show the path from the continuum to quantum mechanics:

1. Vortex structures W_n exist in the continuum and represent electrons.
2. Their sub-geometries c_i form a shared coupling structure $\mathcal{K}_{12}$.
3. This structure is translated by the operator $\hat{P}_{\text{QM}}$ into the observable quantum state $|\Psi_{12}\rangle$.

Thus, the entanglement of two electrons is not simply assumed but *derived from the geometry of the continuum*. This is the central idea of the Monistic Continuum Model: quantum phenomena arise from the structure and dynamics of a continuous physical substrate.

2 Introduction

This section addresses the question: Why is a continuum-based reformulation of electron entanglement necessary?

Electron entanglement is conventionally formulated within the linear Hilbert-space framework of quantum mechanics, where composite systems are represented through tensor-product structures and nonlocal correlations arise from the algebraic properties of state vectors. Although this formulation is operationally successful, it leaves several conceptual and structural questions unresolved: the ontological status of the wave function, the origin of nonlocal correlations, the physical meaning of tensor-product composition, and the absence of a geometric mechanism underlying entanglement. These issues motivate the search for a deeper, continuum-based reformulation.

In the Monistic Continuum Model (MCM), electrons are not point-like particles but stable, vortex-like excitations of a continuous substrate

$$C : M \times \Lambda_\Omega \rightarrow V,$$

where M is a differentiable manifold, V an internal state space, and Λ_Ω an evolution parameter. The continuum possesses a correlation kernel

$$K(x, y; \Lambda_\Omega) = \langle C(x, \Lambda_\Omega), C(y, \Lambda_\Omega) \rangle_V,$$

which encodes both local and nonlocal structure. The organization of the continuum is governed by an operator

$$D_\Omega = D_\alpha + D_{\text{nl}},$$

where D_α stabilizes vortex modes and D_{nl} generates nonlocal coherence. This decomposition provides a geometric and dynamical foundation for phenomena that appear merely as algebraic constructs in standard quantum mechanics.

A central motivation for the continuum approach is the need for a physically grounded mechanism for entanglement. In the MCM, entanglement arises when two electron modes M_1 and M_2 share a non-separable continuum region encoded by the nonlocal term D_{nl} , thereby producing a joint configuration

$$C = M_1 + M_2 + A_C,$$

where A_C is a coherence field induced by D_{nl} . This formulation replaces the abstract tensor-product structure with a geometric non-separability condition,

$$K(x_1, x_2; \Lambda_\Omega) \neq K(x_1; \Lambda_\Omega) K(x_2; \Lambda_\Omega),$$

which directly expresses continuum correlation.

The continuum perspective also provides a natural pathway to the emergence of Hilbert-space structures. Coarse-graining over Λ_Ω yields an effective linear operator

$$\mathcal{H} = \int D_\Omega w(\Lambda_\Omega) d\Lambda_\Omega,$$

whose spectral decomposition gives rise to superposition, linearity, and tensor-product composition. Thus, quantum mechanics appears as an emergent theory derived from deeper continuum dynamics. Comparisons with quantum field theory (QFT) highlight further motivations. QFT treats fields as operator-valued distributions but provides no geometric mechanism for nonlocal correlations. In contrast, the MCM attributes nonlocality to the structure of the correlation kernel K and the operator D_{nl} , offering a physically interpretable origin of entanglement. Similarly, general relativity (GR) describes the geometry of spacetime but does not incorporate nonlocal quantum correlations; the MCM suggests a unified geometric substrate capable of supporting both local curvature and nonlocal coherence.

A continuum-based reformulation is therefore necessary to address fundamental gaps in standard quantum mechanics: the origin of nonlocality, the physical meaning of entanglement, the emergence of Hilbert-space structures, and the geometric interpretation of quantum coherence. By grounding entanglement in continuum geometry and dynamics, the MCM provides a coherent and physically motivated framework that extends and deepens the conceptual foundations of quantum mechanics.

3 Related Work

This section examines the question: How does the MCM relate to existing nonlocal, geometric, and relational quantum-theoretical approaches?

The Monistic Continuum Model (MCM) proposes that electrons are stable, vortex-like excitations of a continuous substrate

$$C : M \times \Lambda_\Omega \rightarrow V,$$

whose organization is governed by an operator

$$D_{\Omega} = D_{\alpha} + D_{\text{nl}},$$

and whose correlations are encoded in a kernel

$$K(x, y; \Lambda_{\Omega}) = \langle C(x, \Lambda_{\Omega}), C(y, \Lambda_{\Omega}) \rangle_V.$$

To situate this framework within existing approaches to nonlocality and geometry in quantum theory, we compare it with standard quantum mechanics (QM), quantum field theory (QFT), geometric and topological formulations, and relational quantum-theoretical approaches.

3.1 Ontological Layers of Existing Frameworks

A structural distinction clarifies how the MCM relates to other approaches. Standard QM and QFT operate on an *algebraic ontology*: states are elements of Hilbert spaces, and correlations arise from operator algebra. Geometric and topological approaches introduce manifold structures but retain the primacy of the Hilbert space. Relational approaches reinterpret correlations but do not provide a physical substrate. General relativity (GR) is geometric but not quantum. The MCM differs by combining a *geometric ontology* with *nonlocal operator organization*, placing it uniquely among these frameworks.

3.2 Standard Quantum Mechanics (QM)

In conventional QM, entanglement arises from the tensor-product structure of Hilbert spaces and the linearity of unitary evolution. Composite systems are represented by states $\psi \in \mathcal{H}_1 \otimes \mathcal{H}_2$, and nonlocal correlations are encoded in the algebraic non-separability of ψ . QM provides no geometric mechanism for nonlocality; the wave function exists in an abstract state space rather than in a physical continuum. In contrast, the MCM anchors nonlocality in the structure of the correlation kernel K and the nonlocal component D_{nl} , with entanglement arising from non-separable continuum configurations rather than purely algebraic tensor products.

3.3 Quantum Field Theory (QFT)

QFT extends QM by elevating fields to operator-valued distributions on spacetime, with locality enforced through commutation relations and microcausality. Although QFT successfully describes particle creation, annihilation, and interactions, its treatment of entanglement remains tied to Hilbert-space factorization and region-based algebras. The MCM differs by treating the continuum $C(x, \Lambda_{\Omega})$ as a physically real substrate whose geometric and topological properties directly generate

electron modes and their correlations. Nonlocality is encoded in the kernel K and the operator D_{nl} , not solely in the algebra of local field operators.

3.4 Geometric and Topological Quantum Theories

Geometric and topological approaches to quantum theory—including fiber-bundle formulations, geometric quantization, and topological quantum field theories—introduce manifolds, connections, and invariants to describe quantum states and phases. These approaches emphasize holonomies, Berry phases, and topological protection but typically retain the Hilbert-space ontology. The MCM shares the geometric emphasis—electron modes are vortex-like structures with topological invariants—but goes further by treating the continuum as ontologically primary and deriving Hilbert-space structures as emergent from coarse-grained organization and spectral properties of D_Ω .

3.5 Relational and Nonlocal Approaches

Relational and nonlocal approaches, such as relational quantum mechanics, consistent histories, and certain nonlocal hidden-variable models, focus on correlations and relations rather than absolute states. They reinterpret the meaning of the wave function and measurement but often lack a concrete geometric substrate supporting these relations. In the MCM, relations between modes M_1 and M_2 are realized as shared continuum regions and non-separable kernel structures:

$$K(x_1, x_2; \Lambda_\Omega) \neq K(x_1; \Lambda_\Omega) K(x_2; \Lambda_\Omega),$$

providing a physically interpretable basis for relational properties.

3.6 General Relativity (GR)

From the perspective of general relativity, spacetime geometry is central, yet quantum nonlocality is not integrated into the classical manifold structure. The MCM proposes a unified geometric substrate in which both local curvature and nonlocal coherence can be encoded in the continuum field C and its kernel K . This positions the MCM as a potential framework bridging geometric and quantum aspects of reality.

3.7 Summary

The MCM relates to existing nonlocal, geometric, and relational quantum-theoretical approaches by sharing their emphasis on geometry, topology, and correlation, while distinguishing itself

through its ontological commitment to a continuous substrate and its derivation of entanglement from non-separable continuum configurations. This comparison motivates the subsequent formal development of the MCM and highlights its role as a continuum-based extension of standard quantum mechanics.

4 Mathematical Foundations of the MCM

This section addresses the question: What is the precise mathematical structure of the continuum substrate $C(x, \Lambda_\Omega)$?

In the Monistic Continuum Model (MCM), the fundamental physical entity is not a discrete particle but a continuous substrate. This substrate is represented by a field

$$C : M \times \Lambda_\Omega \rightarrow V,$$

where M is a smooth, connected, oriented n -dimensional manifold (typically $n = 3$ for physical space or $n = 4$ for spacetime), $\Lambda_\Omega \subseteq \mathbb{R}$ is an evolution-parameter domain, and V is a finite-dimensional internal state space equipped with an inner product $\langle \cdot, \cdot \rangle_V$. The pair (M, V) defines the geometric and internal degrees of freedom of the continuum.

4.1 Continuum Configuration Space and Function Spaces

The configuration space of the continuum is defined as

$$\mathcal{C} = \{C \mid C \in \mathcal{F}(M \times \Lambda_\Omega, V)\},$$

where $\mathcal{F}(M \times \Lambda_\Omega, V)$ denotes an appropriate function space, such as $C^\infty(M \times \Lambda_\Omega, V)$ or a Sobolev space $H^k(M \times \Lambda_\Omega, V)$, depending on regularity requirements. For fixed Λ_Ω , the instantaneous configuration is

$$C_{\Lambda_\Omega} : M \rightarrow V, \quad C_{\Lambda_\Omega}(x) := C(x, \Lambda_\Omega),$$

and the space of instantaneous configurations is

$$\mathcal{C}_{\Lambda_\Omega} = \{C_{\Lambda_\Omega} \mid C_{\Lambda_\Omega} \in \mathcal{F}(M, V)\}.$$

We equip $\mathcal{C}_{\Lambda_\Omega}$ with an inner product

$$\langle C_{\Lambda_\Omega}, C'_{\Lambda_\Omega} \rangle_{\mathcal{C}} = \int_M \langle C_{\Lambda_\Omega}(x), C'_{\Lambda_\Omega}(x) \rangle_V d\mu_C(x),$$

where $d\mu_C$ is a measure induced by the metric g on M . Under suitable completeness assumptions, $\mathcal{C}_{\Lambda_\Omega}$ becomes a Hilbert space, providing a natural bridge to standard quantum systems.

4.2 Correlation Kernel and Nonlocal Structure

The continuum substrate possesses a correlation kernel

$$K : M \times M \times \Lambda_\Omega \rightarrow \mathbb{C}, \quad K(x, y; \Lambda_\Omega) = \langle C(x, \Lambda_\Omega), C(y, \Lambda_\Omega) \rangle_V,$$

which encodes both local and nonlocal correlations. For each fixed Λ_Ω , the kernel $K_{\Lambda_\Omega}(x, y)$ defines an integral operator

$$(\mathcal{K}_{\Lambda_\Omega} f)(x) = \int_M K_{\Lambda_\Omega}(x, y) f(y) d\mu_C(y),$$

acting on suitable function spaces over M . The kernel is *separable* if it factorizes as

$$K_{\Lambda_\Omega}(x, y) = k_{\Lambda_\Omega}(x) k_{\Lambda_\Omega}(y),$$

and *non-separable* otherwise. Non-separability of K_{Λ_Ω} expresses continuum entanglement: correlations cannot be reduced to products of single-point contributions.

4.3 Dynamical Operator and Decomposition

The organization of the continuum is governed by a dynamical operator

$$D_\Omega = D_\alpha + D_{\text{nl}},$$

acting on $\mathcal{C}$. The local part D_α is typically a differential operator constructed from the metric g and possibly additional structures such as connections or potentials, e.g.

$$D_\alpha = a_1 \Delta_g + a_2 \nabla_g + a_3,$$

where Δ_g is the Laplace–Beltrami operator and ∇_g a covariant derivative. The nonlocal part D_{nl} is defined via a kernel $\mathcal{J}$,

$$(D_{\text{nl}} C)(x, \Lambda_\Omega) = \int_M \mathcal{J}(x, y; \Lambda_\Omega) C(y, \Lambda_\Omega) d\mu_C(y),$$

where $\mathcal{J}$ is related to K . The full organization is given by the equation

$$\nabla_{\Lambda_\Omega} C(x, \Lambda_\Omega) = D_\Omega C(x, \Lambda_\Omega),$$

which extends local field equations by nonlocal coherence.

4.4 Electron Modes as Vortex Solutions

Electron modes are defined as stable, localized solutions $M_i(x, \Lambda_\Omega)$ of the continuum organization satisfying

$$D_\Omega M_i = 0,$$

or more generally

$$\nabla_{\Lambda_\Omega} M_i = D_\Omega M_i,$$

under appropriate stability conditions. These modes possess topological and geometric invariants—such as phase winding numbers, circulation integrals, and energy functionals—that correspond to physical quantities like spin, charge, and mass. Mathematically, they can be modeled as vortex-like structures in C_{Λ_Ω} , e.g., as zeros of C_{Λ_Ω} with nontrivial phase winding in V .

4.5 Emergent Hilbert-Space Structure

Under coarse-graining over Λ_Ω , we define an effective operator

$$\mathcal{H} = \int_{\Lambda_\Omega} w(\Lambda_\Omega) D_{\Lambda_\Omega} d\Lambda_\Omega,$$

where $w(\Lambda_\Omega)$ is a weight function and D_{Λ_Ω} the restriction of D_Ω at parameter Λ_Ω . The spectral decomposition of $\mathcal{H}$ yields eigenmodes

$$\mathcal{H}\phi_n = E_n\phi_n,$$

which form a basis of an emergent Hilbert space $\mathcal{H}_{\text{eff}}$. Superposition, linearity, and tensor-product composition arise as effective structures derived from continuum dynamics rather than as fundamental postulates.

4.6 Comparison with QM, QFT, and GR

In standard QM, the primary structure is a Hilbert space $\mathcal{H}$ of state vectors whose dynamics are governed by a Hamiltonian operator H . The MCM, by contrast, begins with a continuum field C on a manifold M and derives an effective Hilbert space from its organization. In QFT, fields are operator-valued distributions on spacetime, but nonlocal correlations are encoded in algebraic

structures rather than in a correlation kernel of a physically real continuum. The MCM introduces a concrete kernel K and a nonlocal operator D_{nl} as generators of coherence. GR provides a geometric manifold with metric g but does not incorporate quantum nonlocality; the MCM extends this geometric structure by adding an internal state space V and a continuum field C capable of supporting both local curvature and nonlocal coherence.

4.7 Result: Precise Mathematical Structure of the Continuum

The continuum substrate $C(x, \Lambda_\Omega)$ is thus specified as a field on a smooth manifold M with values in an internal state space V , equipped with a correlation kernel K , a dynamical operator $D_\Omega = D_\alpha + D_{\text{nl}}$, and a function-space structure that admits an emergent Hilbert space. This mathematical formulation provides the foundation for treating electron entanglement as a continuum-geometric phenomenon and for deriving quantum-like structures from deeper continuum dynamics.

5 Electron Modes as Vortex Solutions

This section examines the question: How do stable vortex modes arise as electron configurations in the continuum?

In the Monistic Continuum Model (MCM), electrons are not modeled as point-like particles but as stable, localized vortex solutions of the continuum field

$$C : M \times \Lambda_\Omega \rightarrow V,$$

where M is a smooth manifold, $\Lambda_\Omega \subseteq \mathbb{R}$ an evolution-parameter domain, and V an internal state space. The goal of this section is to provide a precise mathematical characterization of such vortex modes, identify their invariants, and show how they correspond to electron configurations.

Vortex Configurations in the Continuum

For fixed Λ_Ω , we consider the instantaneous field

$$C_{\Lambda_\Omega} : M \rightarrow V, \quad C_{\Lambda_\Omega}(x) := C(x, \Lambda_\Omega).$$

A vortex configuration is a localized region in M where C_{Λ_Ω} exhibits nontrivial phase structure or circulation. Let γ be a closed loop in M enclosing a region where C_{Λ_Ω} vanishes or becomes singular. We define the phase map

$$\theta_{\Lambda_\Omega} : M \setminus Z_{\Lambda_\Omega} \rightarrow S^1,$$

where Z_{Λ_Ω} is the set of zeros or singularities of C_{Λ_Ω} , and the winding number

$$w(\gamma; \Lambda_\Omega) = \frac{1}{2\pi} \oint_\gamma d\theta_{\Lambda_\Omega}.$$

A nonzero integer $w(\gamma; \Lambda_\Omega)$ indicates a vortex-like structure. Electron modes are identified with vortex configurations that are stable under the organization generated by the operator D_Ω .

Dynamical Equations and Stability

The continuum organization is governed by

$$\nabla_{\Lambda_\Omega} C(x, \Lambda_\Omega) = D_\Omega C(x, \Lambda_\Omega), \quad D_\Omega = D_\alpha + D_{nl}.$$

We seek solutions $M_i(x, \Lambda_\Omega)$ that are localized, vortex-like, and stable. A stationary vortex mode satisfies

$$\nabla_{\Lambda_\Omega} M_i(x, \Lambda_\Omega) = 0,$$

which implies

$$D_\Omega M_i = 0.$$

More generally, a dynamically stable mode satisfies

$$\nabla_{\Lambda_\Omega} M_i = D_\Omega M_i,$$

under stability conditions such as bounded energy and preserved topological invariants. We define an energy functional

$$E[M_i] = \int_M \mathcal{E}(M_i(x, \Lambda_\Omega), \nabla M_i(x, \Lambda_\Omega)) d\mu_C(x),$$

where $\mathcal{E}$ is a local energy density depending on C and its derivatives. Stability requires that $E[M_i]$ is minimized (or stationary) under variations of M_i within an appropriate function space, subject to topological constraints.

Topological and Geometric Invariants

Electron modes as vortex solutions possess invariants corresponding to physical quantities. The winding number w is associated with spin-like properties, while circulation integrals

$$\Gamma(\gamma; \Lambda_\Omega) = \oint_\gamma \mathbf{j}_{\Lambda_\Omega} \cdot d\ell,$$

where $\mathbf{j}_{\Lambda_\Omega}$ is a current derived from C_{Λ_Ω} , can be related to charge-like properties. Mass-like properties can be associated with energy functionals and localization scales. These invariants are robust under continuous deformations of the field that do not cross singularities, providing topological protection for electron modes.

Function Spaces and Mode Classification

We consider the space of admissible vortex configurations

$$\mathcal{M}_{\text{vortex}} = \{M_i \mid M_i \in \mathcal{F}(M \times \Lambda_\Omega, V), D_\Omega M_i = 0, \text{vortex conditions}\},$$

where vortex conditions include nontrivial winding numbers, localization, and finite energy. This space can be decomposed into sectors labeled by topological invariants, e.g.

$$\mathcal{M}_{\text{vortex}} = \bigoplus_{w \in \mathbb{Z}} \mathcal{M}_w,$$

where $\mathcal{M}_w$ contains modes with winding number w . Electron configurations correspond to specific sectors, such as $w = \pm 1$, with additional constraints on energy and localization.

Two-Mode Configurations and Interaction

When two electron modes M_1 and M_2 are present, the total field can be written as

$$C(x, \Lambda_\Omega) = M_1(x, \Lambda_\Omega) + M_2(x, \Lambda_\Omega) + A_C(x, \Lambda_\Omega),$$

where A_C is a coherence field induced by the nonlocal operator D_{nl} . The interaction between modes is mediated by the continuum and encoded in the correlation kernel

$$K(x, y; \Lambda_\Omega) = \langle C(x, \Lambda_\Omega), C(y, \Lambda_\Omega) \rangle_V.$$

Non-separability of K in regions where both modes are present indicates entanglement-like correlations at the continuum level.

Comparison with QM, QFT, and GR

In standard QM, electrons are described by wave functions $\psi(x)$ in a Hilbert space, without explicit vortex structure in physical space unless additional constructs such as phase singularities are introduced. In QFT, electrons are excitations of a fermionic field, but the field is operator-valued and the notion of a classical vortex is not fundamental. The MCM, by contrast, treats electron configurations as vortex solutions of a physically real continuum field, with topological invariants corresponding to physical properties. GR provides a geometric manifold but does not describe matter as vortex modes of a continuum; the MCM adds an internal field structure capable of supporting such configurations.

Result: Electron Modes as Stable Vortex Solutions

Electron configurations in the MCM can be modeled as stable vortex solutions of the continuum field $C(x, \Lambda_\Omega)$, characterized by topological and geometric invariants, governed by a dynamical operator D_Ω , and classified within appropriate function spaces. These vortex modes provide a concrete, continuum-based realization of electrons and form the foundation for the subsequent analysis of entanglement as a continuum-geometric phenomenon.

Vortex Configurations in the Continuum

For fixed $\lambda \in \Lambda$, we consider the instantaneous field

$$C_\lambda : M \rightarrow V, \quad C_\lambda(x) := C(x, \lambda).$$

A vortex configuration is a localized region in M where C_λ exhibits nontrivial phase structure or circulation. Let γ be a closed loop in M enclosing a region where C_λ vanishes or is singular in an appropriate sense. We define the phase map

$$\theta_\lambda : M \setminus Z_\lambda \rightarrow S^1,$$

where Z_λ is the set of zeros or singularities of C_λ , and the winding number

$$w(\gamma; \lambda) = \frac{1}{2\pi} \oint_\gamma d\theta_\lambda.$$

A nonzero integer $w(\gamma; \lambda)$ indicates a vortex-like structure. Electron modes are identified with vortex configurations that are stable under the dynamics generated by the operator D .

Dynamical Equations and Stability

The continuum dynamics are governed by

$$\frac{\partial}{\partial \lambda} C(x, \lambda) = D C(x, \lambda),$$

with

$$D = D_{\text{loc}} + D_{\text{nl}}.$$

We seek solutions $M_i(x, \lambda)$ that are localized, vortex-like, and stable. A stationary vortex mode satisfies

$$\frac{\partial}{\partial \lambda} M_i(x, \lambda) = 0,$$

which implies

$$D M_i = 0.$$

More generally, a dynamically stable mode satisfies

$$\frac{\partial}{\partial \lambda} M_i = D M_i,$$

under stability conditions such as bounded energy and preserved topological invariants. We define an energy functional

$$E[M_i] = \int_M \mathcal{E}(M_i(x, \lambda), \nabla M_i(x, \lambda)) d\mu(x),$$

where $\mathcal{E}$ is a local energy density depending on C and its derivatives. Stability requires that $E[M_i]$ is minimized (or at least stationary) under variations of M_i within an appropriate function space, subject to topological constraints.

Topological and Geometric Invariants

Electron modes as vortex solutions possess invariants corresponding to physical quantities. The winding number w is associated with spin-like properties, while circulation integrals

$$\Gamma(\gamma; \lambda) = \oint_{\gamma} \mathbf{j}_{\lambda} \cdot d\ell,$$

where $\mathbf{j}_{\lambda}$ is a current derived from C_{λ} , can be related to charge-like properties. Mass-like properties can be associated with energy functionals and localization scales. These invariants are robust

under continuous deformations of the field that do not cross singularities, providing topological protection for electron modes.

Function Spaces and Mode Classification

We consider the space of admissible vortex configurations

$$\mathcal{M}_{\text{vortex}} = \{M_i \mid M_i \in \mathcal{F}(M \times \Lambda, V), \, DM_i = 0, \text{ vortex conditions}\},$$

where vortex conditions include nontrivial winding numbers, localization, and finite energy. This space can be decomposed into sectors labeled by topological invariants, e.g.

$$\mathcal{M}_{\text{vortex}} = \bigoplus_{w \in \mathbb{Z}} \mathcal{M}_w,$$

where $\mathcal{M}_w$ contains modes with winding number w . Electron configurations correspond to specific sectors, such as $w = \pm 1$, with additional constraints on energy and localization.

Two-Mode Configurations and Interaction

When two electron modes M_1 and M_2 are present, the total field can be written as

$$C(x, \lambda) = M_1(x, \lambda) + M_2(x, \lambda) + A_C(x, \lambda),$$

where A_C is a coherence field induced by the nonlocal operator D_{nl} . The interaction between modes is mediated by the continuum and encoded in the correlation kernel

$$K(x, y; \lambda) = \langle C(x, \lambda), C(y, \lambda) \rangle_V.$$

Non-separability of K in regions where both modes are present indicates entanglement-like correlations at the continuum level.

Comparison with QM, QFT, and GR

In standard QM, electrons are described by wave functions $\psi(x)$ in a Hilbert space, without explicit vortex structure in physical space unless additional constructs such as phase singularities are introduced. In QFT, electrons are excitations of a fermionic field, but the field is operator-valued and the notion of a classical vortex is not fundamental. The MCM, by contrast, treats electron configurations as vortex solutions of a physically real continuum field, with topological invariants corresponding to physical properties. GR provides a geometric manifold but does not describe

matter as vortex modes of a continuum; the MCM adds an internal field structure capable of supporting such configurations.

Result: Electron Modes as Stable Vortex Solutions

Electron configurations in the MCM can be modeled as stable vortex solutions of the continuum field $C(x, \lambda)$, characterized by topological and geometric invariants, governed by a dynamical operator D , and classified within appropriate function spaces. These vortex modes provide a concrete, continuum-based realization of electrons and form the foundation for the subsequent analysis of entanglement as a continuum-geometric phenomenon.

Vortex Modes in the Continuum (Short Version)

This section examines how stable vortex modes arise as electron configurations in the continuum.

In the Monistic Continuum Model (MCM), electrons are not modeled as point-like particles but as stable, localized vortex solutions of the continuum field

$$C : M \times \Lambda \rightarrow V,$$

where M is a smooth manifold, $\Lambda \subseteq \mathbb{R}$ an evolution-parameter domain, and V an internal state space. The goal is to provide a precise mathematical characterization of such vortex modes, identify their invariants, and show how they correspond to electron configurations.

6 Emergence of Quantum Mechanics

This section addresses the question: How do Hilbert-space structures and linear quantum mechanics emerge from continuum dynamics?

In the Monistic Continuum Model (MCM), the fundamental entity is the continuum field

$$C : M \times \Lambda_\Omega \rightarrow V,$$

whose organization is governed by

$$\nabla_{\Lambda_\Omega} C(x, \Lambda_\Omega) = D_\Omega C(x, \Lambda_\Omega), \quad D_\Omega = D_\alpha + D_{\text{nl}}.$$

Quantum mechanics, by contrast, begins with a Hilbert space $\mathcal{H}$ of state vectors and a Hamiltonian operator H . The goal of this section is to show how Hilbert-space structures and linear quantum

dynamics can emerge from the deeper continuum organization of C .

6.1 Instantaneous Configuration Space and Inner Product

For fixed Λ_Ω , the instantaneous continuum configuration is

$$C_{\Lambda_\Omega} : M \rightarrow V, \quad C_{\Lambda_\Omega}(x) := C(x, \Lambda_\Omega).$$

We define the space of admissible configurations

$$\mathcal{C}_{\Lambda_\Omega} = \{C_{\Lambda_\Omega} \mid C_{\Lambda_\Omega} \in \mathcal{F}(M, V)\},$$

equipped with the CCT inner product

$$\langle C_{\Lambda_\Omega}, C'_{\Lambda_\Omega} \rangle_{\mathcal{C}} = \int_M \langle C_{\Lambda_\Omega}(x), C'_{\Lambda_\Omega}(x) \rangle_V d\mu_C(x),$$

where $d\mu_C$ is induced by the metric g on M . Under suitable completeness and regularity conditions, $\mathcal{C}_{\Lambda_\Omega}$ becomes a Hilbert space. This is the first step toward an emergent quantum structure: continuum configurations at fixed Λ_Ω form a Hilbert space of states.

6.2 Correlation Kernel and Effective Operators

The correlation kernel

$$K(x, y; \Lambda_\Omega) = \langle C(x, \Lambda_\Omega), C(y, \Lambda_\Omega) \rangle_V$$

defines an integral operator

$$(\mathcal{K}_{\Lambda_\Omega} f)(x) = \int_M K(x, y; \Lambda_\Omega) f(y) d\mu_C(y),$$

acting on functions $f : M \rightarrow \mathbb{C}$. Under appropriate conditions (symmetry, positivity, compactness), $\mathcal{K}_{\Lambda_\Omega}$ admits a spectral decomposition

$$\mathcal{K}_{\Lambda_\Omega} \phi_n(x; \Lambda_\Omega) = \kappa_n(\Lambda_\Omega) \phi_n(x; \Lambda_\Omega),$$

where the eigenfunctions ϕ_n form an orthonormal basis of a Hilbert space $\mathcal{H}_{\Lambda_\Omega}$. These eigenfunctions can be interpreted as emergent quantum states, and the kernel $\mathcal{K}_{\Lambda_\Omega}$ as an effective correlation operator.

6.3 Coarse-Graining over the Evolution Parameter

To obtain a time-like evolution analogous to quantum mechanics, we introduce coarse-graining over Λ_Ω . Let $w(\Lambda_\Omega)$ be a weight function and define

$$\mathcal{H}_{\text{eff}} = \int_{\Lambda_\Omega} w(\Lambda_\Omega) \mathcal{H}_{\Lambda_\Omega} d\Lambda_\Omega,$$

in the sense of a direct integral or suitable averaging procedure. Similarly, we define an effective operator

$$\mathcal{H} = \int_{\Lambda_\Omega} w(\Lambda_\Omega) D_{\Lambda_\Omega} d\Lambda_\Omega,$$

where D_{Λ_Ω} is the restriction of D_Ω at parameter Λ_Ω . The operator $\mathcal{H}$ acts on $\mathcal{H}_{\text{eff}}$ and plays the role of an emergent Hamiltonian.

6.4 Spectral Decomposition and Linear Dynamics

Assuming that the operator $\mathcal{H}$ is self-adjoint (or suitably extendable), we obtain a spectral decomposition

$$\mathcal{H}\psi_n = E_n\psi_n,$$

with eigenfunctions $\psi_n \in \mathcal{H}_{\text{eff}}$ forming an orthonormal basis. Any state $\Psi \in \mathcal{H}_{\text{eff}}$ can be expanded as

$$\Psi = \sum_n c_n \psi_n,$$

with complex coefficients c_n . The organization of Ψ under continuum dynamics induces an effective linear evolution

$$i \frac{d}{dt} \Psi(t) = \mathcal{H} \Psi(t),$$

where t is an emergent time parameter arising from coarse-graining over Λ_Ω . This equation is formally identical to the Schrödinger equation but derived from continuum dynamics rather than postulated.

6.5 Tensor-Product Structures and Composite Systems

Composite systems in quantum mechanics are described by tensor products $\mathcal{H}_1 \otimes \mathcal{H}_2$. In the MCM, composite configurations arise from multiple localized modes $M_1, M_2, \dots$ within the continuum

field C . The total field can be written as

$$C(x, \Lambda_\Omega) = \sum_i M_i(x, \Lambda_\Omega) + A_C(x, \Lambda_\Omega),$$

where A_C is a coherence field induced by D_{nl} . The correlation kernel

$$K(x, y; \Lambda_\Omega) = \langle C(x, \Lambda_\Omega), C(y, \Lambda_\Omega) \rangle_V$$

can be decomposed into contributions associated with individual modes and their interactions. Under suitable approximations (e.g., weak overlap, localized modes), the effective Hilbert space $\mathcal{H}_{\text{eff}}$ factorizes into subspaces associated with each mode, yielding a tensor-product structure. Entanglement corresponds to non-factorizable kernel structures.

Comparison with QM, QFT, and GR

In standard QM, Hilbert spaces and linear dynamics are fundamental postulates. In the MCM, they emerge from the continuum field C , its correlation kernel K , and the dynamical operator D_Ω . QFT introduces fields on spacetime and operator algebras but still relies on Hilbert spaces as primary structures. The MCM provides a deeper level in which Hilbert spaces arise from the spectral properties of continuum operators. GR describes spacetime geometry via a metric g but does not include quantum-dynamical structures; the MCM suggests that quantum structures can emerge from a continuum living on a geometric manifold.

Result: Emergence of Hilbert-Space Quantum Mechanics

Starting from the continuum field $C(x, \Lambda_\Omega)$, its correlation kernel K , and the dynamical operator D_Ω , one can construct Hilbert spaces of configurations, define effective operators through coarse-graining, and derive linear dynamics formally equivalent to the Schrödinger equation. Tensor-product structures and entanglement emerge from multi-mode configurations and non-separable kernels. Thus, Hilbert-space quantum mechanics appears as an emergent, effective theory arising from deeper continuum dynamics in the MCM.

7 Two-Mode Continuum Configurations

This section examines the question: Under which conditions does a two-mode continuum configuration in the timeless continuum become non-separable?

In the Monistic Continuum Model (MCM), electron configurations arise as vortex-like modes of the continuum field

$$C : M \times \Lambda_\Omega \rightarrow V,$$

where Λ_Ω denotes the projective organizational dimension of the medium. The dynamics are not temporal but determined by stress and organizational gradients:

$$\nabla_{\Lambda_\Omega} C(x, \Lambda_\Omega) = D_\Omega C(x, \Lambda_\Omega), \quad D_\Omega = D_\alpha + D_{\text{nl}},$$

where D_α represents the local stress operator and D_{nl} the nonlocal coherence operator. When two stable vortex modes M_1 and M_2 coexist in the continuum, the total configuration becomes

$$C(x, \Lambda_\Omega) = M_1(x, \Lambda_\Omega) + M_2(x, \Lambda_\Omega) + A_C(x, \Lambda_\Omega),$$

where A_C is a coherence field induced by D_{nl} . The central question is: Under which ontological–axiometric conditions does such a two-mode configuration become non-separable?

7.1 Two-Mode Configuration Space

For fixed Λ_Ω , the instantaneous configuration space is

$$\mathcal{C}_{\Lambda_\Omega} = \mathcal{F}(M, V),$$

equipped with the CCT inner product

$$\langle C_{\Lambda_\Omega}, C'_{\Lambda_\Omega} \rangle_c = \int_M \langle C_{\Lambda_\Omega}(x), C'_{\Lambda_\Omega}(x) \rangle_V d\mu_C(x),$$

where $d\mu_C$ denotes the CCT measure of the medium. The two-mode subspace is

$$\mathcal{C}_{\Lambda_\Omega}^{(2)} = \{M_1 + M_2 + A_C \mid M_i \in \mathcal{M}_{\text{vortex}}, A_C \in \mathcal{F}(M, V)\}.$$

A configuration is *separable* if it can be written as

$$C(x, \Lambda_\Omega) = C_1(x, \Lambda_\Omega) + C_2(x, \Lambda_\Omega),$$

where C_1 and C_2 possess disjoint stress domains or are factorizable. Otherwise, it is *non-separable*.

7.2 Correlation Kernel and Separability

The correlation kernel

$$K(x, y; \Lambda_\Omega) = \langle C(x, \Lambda_\Omega), C(y, \Lambda_\Omega) \rangle_V$$

is the central mathematical object for determining separability. For a two-mode configuration,

$$K(x, y; \Lambda_\Omega) = K_{11} + K_{22} + K_{12} + K_A,$$

where K_{ij} are the mode contributions and K_A describes the coherence field.

Definition (Separability). A two-mode configuration is *separable* if

$$K(x, y; \Lambda_\Omega) = K_1 + K_2,$$

i.e., no cross term K_{12} appears. It is *non-separable* if

$$K_{12}(x, y; \Lambda_\Omega) \neq 0$$

on a set of positive CCT measure.

Definition (Strong Non-Separability). A configuration is *strongly non-separable* if

$$K(x_1, x_2; \Lambda_\Omega) \neq K(x_1; \Lambda_\Omega) K(x_2; \Lambda_\Omega),$$

i.e., the kernel cannot be factorized into single-point contributions.

7.3 Operator Conditions for Non-Separability

The nonlocal operator D_{nl} acts via

$$(D_{\text{nl}}C)(x, \Lambda_\Omega) = \int_M \mathcal{J}(x, y; \Lambda_\Omega) C(y, \Lambda_\Omega) d\mu_C(y),$$

where $\mathcal{J}$ is the CCT interaction kernel.

Condition 1 (Nonlocal Overlap). If the stress domains of M_1 and M_2 overlap in a region where $\mathcal{J}(x, y; \Lambda_\Omega) \neq 0$, then $K_{12} \neq 0$.

Condition 2 (Generation of a Coherence Field). If D_{nl} generates a non-vanishing A_C , then $K_A \neq 0$, even for disjoint vortex cores.

Condition 3 (Spectral Coupling). Let $\mathcal{K}_{\Lambda_\Omega}$ be the correlation operator. If its spectrum contains mixed eigenfunctions,

$$\phi_n(x; \Lambda_\Omega) = \alpha_n M_1(x, \Lambda_\Omega) + \beta_n M_2(x, \Lambda_\Omega), \quad \alpha_n, \beta_n \neq 0,$$

then the configuration is non-separable.

7.4 Functional-Analytic Characterization

$$\mathcal{S}_{\Lambda_\Omega} = \{C \in \mathcal{C}_{\Lambda_\Omega}^{(2)} \mid K_{12} = 0\}, \quad \mathcal{N}_{\Lambda_\Omega} = \mathcal{C}_{\Lambda_\Omega}^{(2)} \setminus \mathcal{S}_{\Lambda_\Omega}.$$

Theorem (Criterion for Non-Separability). A two-mode configuration $C = M_1 + M_2 + A_C$ is non-separable if and only if

$$\exists f, g \in \mathcal{F}(M, V) : \langle f, \mathcal{K}_{\Lambda_\Omega} g \rangle_C \neq \langle f, \mathcal{K}_{1, \Lambda_\Omega} g \rangle_C + \langle f, \mathcal{K}_{2, \Lambda_\Omega} g \rangle_C.$$

Proof (Sketch). Cross terms prevent the decomposition of the correlation operator. Separability is equivalent to operator decomposability in the CCT space.

7.5 Comparison with QM, QFT, and GR

In QM, separability corresponds to the factorization of the wave function $\psi(x_1, x_2) = \psi_1(x_1)\psi_2(x_2)$. Non-separability corresponds to entanglement. In QFT, non-separability arises through interactions or entangled fields. In GR, correlations arise geometrically. The MCM provides a geometric mechanism for non-separability via the CCT kernel and nonlocal operators.

7.6 Result: Conditions for Non-Separability

A two-mode continuum configuration is non-separable if and only if:

1. the stress or coherence domains of the two modes overlap in regions where the nonlocal operator D_{nl} acts;
2. D_{nl} generates a shared coherence field A_C that cannot be reduced to individual modes;
3. the associated correlation operator possesses mixed eigenfunctions that cannot be expressed as products of single-mode eigenfunctions;
4. the correlation kernel is structurally non-factorizable, i.e.

$$K(x_1, x_2; \Lambda_\Omega) \neq K_1(x_1; \Lambda_\Omega) K_2(x_2; \Lambda_\Omega).$$

These conditions define the continuum-geometric origin of non-separability: non-separability arises precisely when two localized vortex modes share a region of the continuum in which the nonlocal operator D_{nl} induces coherent organization. In this shared domain, the continuum structure becomes interlinked such that the spectral properties of the correlation operator can no longer be attributed to individual modes.

Formally, this means:

- the kernel contains mixed contributions that cannot be reduced to single-mode terms;
- the operator $\mathcal{K}_{12}$ has at least one eigenfunction $\phi(x_1, x_2)$ that does not admit a product form

$$\phi(x_1, x_2) \neq \phi_1(x_1) \phi_2(x_2);$$

- the coherence field A_C couples the modes through nonlocal integral terms.

Thus, non-separability is a purely geometric–dynamic phenomenon of the continuum: it does not arise from algebraic tensor products but from the overlap of modes, the action of the nonlocal operator, and the resulting shared continuum organization.

This characterization forms the foundation for the subsequent development of entanglement operators in the MCM and shows how quantum correlations emerge from the structure of the continuum.

8 Entanglement Operators in the MCM

This section addresses the question: Which operators generate, preserve, or modify continuum entanglement in the timeless continuum?

In the Monistic Continuum Model (MCM), continuum entanglement arises from the interplay between local stress dynamics and nonlocal coherence mechanisms encoded in the continuum field

$$C : M \times \Lambda_\Omega \rightarrow V,$$

whose organization is governed by

$$\nabla_{\Lambda_\Omega} C(x, \Lambda_\Omega) = D_\Omega C(x, \Lambda_\Omega), \quad D_\Omega = D_\alpha + D_{\text{nl}}.$$

This section identifies and characterizes the operators that generate, preserve, or modify entanglement in the continuum, providing a formal operator-theoretic foundation for non-separable configurations.

8.1 Classes of Operators in the Continuum

Operators acting on the continuum field fall into three categories:

1. **Local operators** D_α : stress operators acting pointwise on C .
2. **Nonlocal operators** D_{nl} : integral operators generating coherence across spatial regions.
3. **Mixed operators** D_{mix} : operators combining local and nonlocal components.

These operators act on the function space

$$\mathcal{C}_{\Lambda_\Omega} = \mathcal{F}(M, V),$$

equipped with the CCT inner product

$$\langle C_{\Lambda_\Omega}, C'_{\Lambda_\Omega} \rangle_C = \int_M \langle C_{\Lambda_\Omega}(x), C'_{\Lambda_\Omega}(x) \rangle_V d\mu_C(x).$$

8.2 Local Operators and Separability

Local operators take the form

$$D_\alpha = a_1 \Delta_g + a_2 \nabla_g + a_3,$$

where Δ_g is the Laplace–Beltrami operator and ∇_g a covariant derivative. Their action is pointwise:

$$(D_\alpha C)(x, \Lambda_\Omega) = F(x, C(x, \Lambda_\Omega), \nabla C(x, \Lambda_\Omega)).$$

Result. Local operators preserve separability. If

$$C = C_1 + C_2,$$

with disjoint stress domains, then

$$D_\alpha C = D_\alpha C_1 + D_\alpha C_2,$$

and no cross terms arise. Local operators therefore cannot generate entanglement.

8.3 Nonlocal Operators and the Generation of Entanglement

Nonlocal operators are defined via an interaction kernel $\mathcal{J}$:

$$(D_{\text{nl}}C)(x, \Lambda_\Omega) = \int_M \mathcal{J}(x, y; \Lambda_\Omega) C(y, \Lambda_\Omega) d\mu_C(y).$$

Definition (Entanglement-generating operator). A nonlocal operator D_{nl} is entanglement-generating if, for a two-mode configuration

$$C = M_1 + M_2,$$

it produces a coherence field

$$A_C = D_{\text{nl}}(M_1 + M_2),$$

such that the correlation kernel

$$K(x, y; \Lambda_\Omega) = \langle C(x, \Lambda_\Omega), C(y, \Lambda_\Omega) \rangle_V$$

becomes non-separable.

Condition (Kernel non-separability). Entanglement is generated when

$$\mathcal{J}(x, y; \Lambda_\Omega) \neq 0 \quad \text{in a region where both } M_1 \text{ and } M_2 \text{ possess stress domains.}$$

Result. Nonlocal operators generate entanglement through cross terms in the correlation kernel:

$$K_{12}(x, y; \Lambda_\Omega) = \langle M_1(x, \Lambda_\Omega), D_{\text{nl}}M_2(y, \Lambda_\Omega) \rangle_V \neq 0.$$

8.4 Mixed Operators and the Modulation of Entanglement

Mixed operators combine local and nonlocal components:

$$D_{\text{mix}} = D_\alpha + \alpha D_{\text{nl}},$$

with $\alpha \in \mathbb{R}$ controlling the strength of nonlocality.

Result. Mixed operators modulate entanglement:

$$\nabla_\alpha K_{12}(x, y; \Lambda_\Omega) = \langle M_1(x, \Lambda_\Omega), D_{\text{nl}}M_2(y, \Lambda_\Omega) \rangle_V.$$

Thus, α acts as a tunable entanglement parameter.

8.5 Spectral Entanglement Operators

The correlation operator

$$(\mathcal{K}_{\Lambda_\Omega} f)(x) = \int_M K(x, y; \Lambda_\Omega) f(y) d\mu_C(y)$$

admits a spectral decomposition

$$\mathcal{K}_{\Lambda_\Omega} \phi_n = \kappa_n \phi_n.$$

Definition (Spectral entanglement operator). An operator $\mathcal{E}$ preserves entanglement if it maps eigenfunctions to eigenfunctions:

$$\mathcal{E} \phi_n = \lambda_n \phi_n.$$

Definition (Spectral disentangler). An operator $\mathcal{B}$ breaks entanglement if it diagonalizes the kernel:

$$\mathcal{B} \mathcal{K}_{\Lambda_\Omega} \mathcal{B}^{-1} \text{ is separable.}$$

Result. Nonlocal operators are spectral entanglement generators; local operators are spectral entanglement preservers.

8.6 Functional-Analytic Characterization

$$\mathcal{N}_{\Lambda_\Omega} = \{C \in \mathcal{C}_{\Lambda_\Omega} \mid K_{12} \neq 0\}, \quad \mathcal{S}_{\Lambda_\Omega} = \{C \in \mathcal{C}_{\Lambda_\Omega} \mid K_{12} = 0\}.$$

Theorem (Operator entanglement criterion). An operator D maps $\mathcal{S}_{\Lambda_\Omega}$ into $\mathcal{N}_{\Lambda_\Omega}$ if and only if

$$\exists f, g \in \mathcal{F}(M, V) : \langle f, Dg \rangle_C \neq \langle f, D_\alpha g \rangle_C.$$

Proof (Sketch). Local operators preserve separability; nonlocal operators generate cross terms. Mixed operators interpolate between both behaviors.

8.7 Comparison with QM, QFT, and GR

In QM, entanglement is generated by nonlocal Hamiltonians or interaction terms. In QFT, entanglement arises from field interactions and vacuum correlations. GR contains no entanglement operators; correlations arise from geometry. The MCM unifies these perspectives by introducing explicit continuous operators that generate both local stress organization and nonlocal coherence. Unlike QM, where non-separability is an algebraic property of the Hilbert space, and unlike QFT, where it arises from operator-valued fields and their commutators, non-separability in the MCM emerges from the structure of the correlation kernel and the action of the nonlocal operator D_{nl} . This operator produces a physically interpretable coherence field A_C that couples the modes across the continuum without requiring a Hilbert space.

While GR describes correlations solely through the metric and geometric curvature, the MCM extends this geometric level by introducing an internal continuum field C , whose nonlocal organization produces entanglement as a geometric coherence phenomenon. Thus, the MCM provides a unified framework in which:

- QM entanglement appears as emergent spectral structure,
- QFT nonlocality appears as field coherence,
- and GR geometry acts as the carrier structure,

all arising from a single continuous substrate.

The MCM therefore shows that entanglement is neither purely algebraic, nor purely field-theoretic, nor purely geometric, but a consequence of the nonlocal organization of a physically real continuum.

9 Main Theorem of the MCM

This section formulates the central question: Can it be shown that continuum coherence generates entanglement equivalent to quantum entanglement?

The Monistic Continuum Model (MCM) postulates that electron entanglement arises from nonlocal coherence within a continuous substrate,

$$C : M \times \Lambda_\Omega \rightarrow V,$$

whose organization is determined by stress and coherence operators:

$$\nabla_{\Lambda_\Omega} C(x, \Lambda_\Omega) = D_\Omega C(x, \Lambda_\Omega), \quad D_\Omega = D_\alpha + D_{nl}.$$

The central question is whether the continuum coherence encoded in the correlation kernel

$$K(x, y; \Lambda_\Omega) = \langle C(x, \Lambda_\Omega), C(y, \Lambda_\Omega) \rangle_V$$

can be shown to generate entanglement equivalent to quantum non-separability. We formulate and prove the Main Theorem of the MCM, which establishes the mathematical conditions under which continuum coherence produces quantum-equivalent entanglement.

9.1 Two-Mode Continuum Configuration

Consider two stable vortex modes M_1 and M_2 of the continuum field. The total configuration is

$$C(x, \Lambda_\Omega) = M_1(x, \Lambda_\Omega) + M_2(x, \Lambda_\Omega) + A_C(x, \Lambda_\Omega),$$

where A_C is a coherence field generated by the nonlocal operator D_{nl} . The correlation kernel decomposes as

$$K = K_{11} + K_{22} + K_{12} + K_A,$$

where K_{12} and K_A contain the coherent cross terms.

Definition (Quantum-Equivalent Entanglement). A continuum configuration exhibits quantum-equivalent entanglement if its correlation kernel is non-separable,

$$K(x_1, x_2; \Lambda_\Omega) \neq K(x_1; \Lambda_\Omega) K(x_2; \Lambda_\Omega),$$

and the induced correlation operator

$$(\mathcal{K}_{\Lambda_\Omega} f)(x) = \int_M K(x, y; \Lambda_\Omega) f(y) d\mu_C(y)$$

possesses mixed eigenfunctions,

$$\phi_n = \alpha_n M_1 + \beta_n M_2, \quad \alpha_n, \beta_n \neq 0.$$

9.2 Operator Conditions for Continuum Coherence

The nonlocal operator acts as

$$(D_{\text{nl}} C)(x, \Lambda_\Omega) = \int_M \mathcal{J}(x, y; \Lambda_\Omega) C(y, \Lambda_\Omega) d\mu_C(y),$$

where $\mathcal{J}$ is the CCT interaction kernel.

Condition 1 (Generation of Coherence). If $\mathcal{J}(x, y; \Lambda_\Omega) \neq 0$ in a region where both modes possess stress domains, then

$$A_C = D_{\text{nl}}(M_1 + M_2) \neq 0.$$

Condition 2 (Kernel Cross-Coupling). If

$$\langle M_1(x, \Lambda_\Omega), D_{\text{nl}} M_2(y, \Lambda_\Omega) \rangle_V \neq 0,$$

then $K_{12} \neq 0$, which implies non-separability.

Condition 3 (Spectral Mixing). If the correlation operator possesses mixed eigenfunctions, the configuration is entangled.

9.3 Functional-Analytic Framework

$$\mathcal{S}_{\Lambda_\Omega} = \{C \in \mathcal{C}_{\Lambda_\Omega} \mid K_{12} = 0\}, \quad \mathcal{N}_{\Lambda_\Omega} = \mathcal{C}_{\Lambda_\Omega} \setminus \mathcal{S}_{\Lambda_\Omega}.$$

Lemma 1 (Operator-Induced Non-Separability). If $D_{\text{nl}} \neq 0$, then in general

$$D_{\text{nl}}(\mathcal{S}_{\Lambda_\Omega}) \subseteq \mathcal{N}_{\Lambda_\Omega}.$$

Proof. For separable $C = C_1 + C_2$,

$$D_{\text{nl}}C = D_{\text{nl}}C_1 + D_{\text{nl}}C_2.$$

Since D_{nl} integrates over all of M , cross terms arise whenever $\mathcal{J}$ does not vanish on the overlap of stress domains. Separability is therefore generically lost.

Theorem (Continuum Coherence Generates Quantum-Equivalent Entanglement). Let $C = M_1 + M_2 + A_C$ be a two-mode continuum configuration. Assume:

1. The nonlocal operator D_{nl} has a kernel $\mathcal{J}$ that does not vanish in a region where both modes possess stress domains.
2. The correlation operator $\mathcal{K}_{\Lambda_\Omega}$ is compact, self-adjoint, and positive.
3. The coherence field A_C generated by D_{nl} is non-vanishing.

Then:

1. The correlation kernel K is non-separable.
2. The correlation operator $\mathcal{K}_{\Lambda_\Omega}$ possesses mixed eigenfunctions.
3. The continuum configuration exhibits quantum-equivalent entanglement.

Proof. (1) Nonlocality of $\mathcal{J}$ implies $K_{12} \neq 0$. (2) Compactness and positivity of $\mathcal{K}_{\Lambda_\Omega}$ guarantee a spectral decomposition. (3) Non-vanishing cross terms generate mixed eigenfunctions. Thus, the configuration is non-separable and satisfies the definition of quantum-equivalent entanglement.

9.4 Comparison with QM, QFT, and GR

In QM, entanglement arises from non-factorizable wave functions. In QFT, entanglement arises from nonlocal correlations of field operators. In GR, no entanglement mechanism exists; correlations arise from geometry.

The MCM provides a geometric mechanism for entanglement through continuum coherence and nonlocal operators, thereby linking quantum nonlocality and geometric structure.

9.5 Result: Formal Proof of Continuum-Induced Entanglement

Continuum coherence, generated by nonlocal operators acting on vortex modes, produces non-separable kernel structures and mixed spectral modes. Under precise mathematical conditions, this coherence leads to entanglement equivalent to quantum entanglement. The Main Theorem demonstrates the continuum-geometric origin of entanglement in the MCM.

10 Predictions and Experimental Signatures

This section investigates the question: Which experimentally accessible signatures distinguish MCM-based entanglement from standard QM?

The Monistic Continuum Model (MCM) predicts that electron entanglement arises from nonlocal coherence within a continuous substrate,

$$C : M \times \Lambda_\Omega \rightarrow V,$$

where correlations are encoded in the kernel

$$K(x, y; \Lambda_\Omega) = \langle C(x, \Lambda_\Omega), C(y, \Lambda_\Omega) \rangle_V$$

and the organization is governed by

$$\nabla_{\Lambda_\Omega} C = D_\Omega C, \quad D_\Omega = D_\alpha + D_{\text{nl}}.$$

This continuum mechanism differs fundamentally from standard quantum mechanics (QM), in which entanglement is an algebraic property of tensor-product Hilbert spaces. The goal of this section is to identify experimentally accessible signatures that distinguish MCM-based entanglement from QM predictions.

Continuum-Induced Correlation Plateaus

Nonlocal coherence arises from the integral operator

$$(D_{\text{nl}}C)(x, \Lambda_\Omega) = \int_M \mathcal{J}(x, y; \Lambda_\Omega) C(y, \Lambda_\Omega) d\mu_C(y),$$

which generates a coherence field A_C . This leads to correlation plateaus in the kernel:

$$K(x, y; \Lambda_\Omega) \approx \text{const.} \quad \text{for } (x, y) \in \Omega_C,$$

where Ω_C is the coherence region.

Prediction 1. Bell-type correlation curves exhibit extended plateaus instead of the sinusoidal dependence

$$E_{\text{QM}}(\theta) = -\cos(\theta),$$

predicted by QM. The plateau width depends on the geometry of Ω_C .

Experimental Signature. High-resolution Bell tests with tunable spatial separation should reveal deviations from the cosine law, especially in regimes where coherence regions overlap.

Geometry-Dependent Entanglement Strength

The continuum kernel depends on the geometry of the underlying manifold M :

$$K(x, y; \Lambda_\Omega) = F(d_g(x, y), \mathcal{J}(x, y; \Lambda_\Omega)),$$

where d_g is the geodesic distance.

Prediction 2. Entanglement strength varies with geometric deformation of the substrate (e.g., curvature, confinement, boundary conditions).

Experimental Signature. Electron interferometry in curved or confined geometries (quantum waveguides, curved graphene structures) should show geometry-dependent modulation of entanglement visibility.

10.1 Kernel-Induced Coherence Modulation

The nonlocal kernel $\mathcal{J}$ generates coherence fields whose strength depends on the overlap of vortex modes:

$$A_C = D_{\text{nl}}(M_1 + M_2).$$

Prediction 3. Entanglement visibility oscillates as a function of mode separation, with coherence maxima at distances of maximal kernel overlap.

Experimental Signature. Two-electron entanglement experiments with tunable spatial separation should show non-monotonic visibility curves, unlike QM's monotonic decoherence behavior.

10.2 Deviation from Exchange Symmetry

In QM, identical fermions obey strict antisymmetry:

$$\psi(x_1, x_2) = -\psi(x_2, x_1).$$

In the MCM, vortex modes possess geometric and topological invariants, and exchange symmetry arises from mode structure rather than algebraic antisymmetry.

Prediction 4. Exchange symmetry violations occur when vortex cores have asymmetric geometry or unequal coherence fields.

Experimental Signature. Electron pair interference experiments should reveal small but measurable deviations from perfect antisymmetry in asymmetric geometries.

10.3 Spectral Signatures of Continuum Entanglement

The correlation operator

$$(\mathcal{K}_{\Lambda\Omega}f)(x) = \int_M K(x, y; \Lambda\Omega) f(y) d\mu_C(y)$$

admits a spectral decomposition

$$\mathcal{K}_{\Lambda\Omega}\phi_n = \kappa_n\phi_n.$$

Prediction 5. Mixed eigenfunctions appear even when vortex cores are spatially separated:

$$\phi_n = \alpha_n M_1 + \beta_n M_2, \quad \alpha_n, \beta_n \neq 0.$$

Experimental Signature. Spectral tomography (e.g., via quantum state reconstruction) should reveal mixed-mode eigenfunctions even in regimes where QM predicts separability.

10.4 Nonlocal Coherence Without Entanglement Collapse

In QM, entanglement collapses upon measurement. In the MCM, coherence fields persist until the continuum configuration is disrupted.

Prediction 6. Post-measurement residual correlations persist longer than QM predicts.

Experimental Signature. Delayed-choice entanglement experiments should show anomalously long-lived correlations.

10.5 Comparison with QM, QFT, and GR

- QM predicts sinusoidal Bell correlations, strict antisymmetry, and collapse upon measurement.
- QFT predicts vacuum-induced correlations but no geometry-dependent modulation of entanglement.
- GR predicts geometric effects on classical fields but no quantum entanglement.

The MCM predicts geometry-dependent entanglement strength, correlation plateaus, mixed spectral modes, and persistent coherence fields—none of which arise naturally in QM, QFT, or GR.

10.6 Result: Experimentally Accessible MCM Signatures

The MCM predicts several experimentally accessible deviations from QM:

1. Correlation plateaus in Bell tests.
2. Geometry-dependent modulation of entanglement.
3. Non-monotonic coherence–separation curves.
4. Deviations from exchange symmetry.
5. Mixed spectral modes in tomography.
6. Persistent post-measurement correlations.

These signatures provide concrete experimental pathways for distinguishing continuum-based entanglement from standard quantum mechanics.

11 Comparison with Standard Quantum Theory

This section addresses the question: How does MCM-based entanglement differ from QM, QFT, GRW, Everett, and relational QM – and how does it reproduce them?

The Monistic Continuum Model (MCM) postulates that electron entanglement arises from nonlocal coherence within a continuous substrate,

$$C : M \times \Lambda_\Omega \rightarrow V,$$

where correlations are encoded in the kernel

$$K(x, y; \Lambda_\Omega) = \langle C(x, \Lambda_\Omega), C(y, \Lambda_\Omega) \rangle_V$$

and the organization is governed by

$$\nabla_{\Lambda_\Omega} C = D_\Omega C, \quad D_\Omega = D_\alpha + D_{\text{nl}}.$$

This section compares MCM-based entanglement with the central frameworks of standard quantum theory: quantum mechanics (QM), quantum field theory (QFT), GRW collapse theory, Everett's many-worlds interpretation, and relational quantum mechanics (RQM). The goal is to identify both the points of reproduction (where MCM yields the same predictions) and the points of deviation (where MCM predicts new phenomena).

11.1 Comparison with Standard Quantum Mechanics (QM)

In QM, entanglement is an algebraic property of tensor-product Hilbert spaces:

$$\psi(x_1, x_2) \in \mathcal{H}_1 \otimes \mathcal{H}_2,$$

with non-separability defined by

$$\psi(x_1, x_2) \neq \psi_1(x_1)\psi_2(x_2).$$

The MCM reproduces QM entanglement through non-separable continuum kernels:

$$K(x_1, x_2; \Lambda_\Omega) \neq K(x_1; \Lambda_\Omega)K(x_2; \Lambda_\Omega),$$

and mixed eigenfunctions of the correlation operator:

$$\mathcal{K}_{\Lambda_\Omega} \phi_n = \kappa_n \phi_n.$$

Reproduction.

- Bell correlations and CHSH violations are reproduced when the coherence region Ω_C is sufficiently large.
- The emergent Hilbert space $\mathcal{H}_{\text{eff}}$ yields linear Schrödinger dynamics.
- Tensor-product structures arise through multi-mode decomposition.

Difference.

- Entanglement arises from continuum geometry, not from algebraic tensor products.
- Correlation plateaus and geometry-dependent modulation are absent in QM.
- Exchange symmetry arises from vortex topology, not from antisymmetrization of wave functions.

11.2 Comparison with Quantum Field Theory (QFT)

QFT describes particles as excitations of operator-valued fields on spacetime. Entanglement arises from nonlocal correlations of field operators and the vacuum structure.

Reproduction.

- MCM reproduces QFT-like nonlocality via the kernel K .
- Vortex modes resemble localized field excitations.
- The continuum field C can be interpreted as a classical precursor to QFT fields.

Difference.

- QFT fields are operator-valued distributions; MCM fields are physically real continuum configurations.
- QFT provides no geometric mechanism for entanglement; MCM does.
- QFT vacuum entanglement is algebraic; MCM coherence is geometric and dynamical.

11.3 Comparison with GRW Collapse Theory

GRW modifies QM through spontaneous collapse events governed by stochastic operators.

Reproduction.

- MCM reproduces GRW-like localization when vortex modes are geometrically perturbed.
- Collapse-like behavior arises when coherence fields A_C are destroyed by boundary conditions or geometric disturbances.

Difference.

- GRW collapse is stochastic and non-geometric; MCM collapse is geometric and dynamical.
- GRW modifies the Schrödinger equation; MCM modifies the continuum operator D_Ω .
- GRW does not predict geometry-dependent collapse rates; MCM does.

11.4 Comparison with Everett's Many-Worlds Interpretation

Everett interprets entanglement as branching of the universal wave function.

Reproduction.

- MCM reproduces Everettian branching as different spectral decompositions of $\mathcal{K}_{\Lambda\Omega}$.
- Mixed eigenfunctions correspond to “branches” in the Everett sense.

Difference.

- Everett requires a universal wave function; MCM requires only a continuum field.
- Branching is geometric in MCM, algebraic in Everett.
- MCM predicts geometry-dependent branching.

11.5 Comparison with Relational Quantum Mechanics (RQM)

RQM interprets quantum states as relations between systems.

Reproduction.

- MCM reproduces relational correlations through non-separable kernel structures.
- Relations correspond to shared continuum regions Ω_C .

Difference.

- RQM has no physical substrate for relations; MCM provides one.
- RQM does not predict geometry-dependent relational strength; MCM does.
- RQM is interpretative; MCM is dynamical and geometric.

11.6 Unified Comparison Table

Framework	Reproduced by MCM	Distinctive MCM Prediction
QM	Bell correlations, Hilbert-space structure	Geometry-dependent entanglement
QFT	Field-like excitations	Real continuum substrate
GRW	Collapse-like localization	Geometric collapse mechanism
Everett	Branching structure	Geometric branching
RQM	Relational correlations	Physical substrate for relations

11.7 Result: MCM Reproduces and Extends Standard Quantum Theory

The MCM reproduces all experimentally confirmed features of QM and QFT while providing:

1. a geometric origin of entanglement,
2. a continuum-based mechanism for coherence,

3. predictions beyond QM (correlation plateaus, geometry-dependent modulation),
4. a unified substrate linking quantum and geometric phenomena.

Thus, MCM entanglement differs in ontology and mechanism from standard quantum theory, while reproducing all empirically verified quantum predictions.

12 Discussion

This section investigates the question: What conceptual and technological implications arise from continuum-based entanglement?

The Monistic Continuum Model (MCM) postulates that entanglement is not an abstract algebraic feature of tensor-product Hilbert spaces, but a geometric and dynamical property of a continuous substrate

$$C : M \times \Lambda_\Omega \rightarrow V,$$

with coherence encoded in the correlation kernel

$$K(x, y; \Lambda_\Omega) = \langle C(x, \Lambda_\Omega), C(y, \Lambda_\Omega) \rangle_V$$

and organized through

$$\nabla_{\Lambda_\Omega} C = D_\Omega C, \quad D_\Omega = D_\alpha + D_{nl}.$$

This section discusses the conceptual, physical, and technological implications of this continuum-based perspective and evaluates how it changes our understanding of entanglement compared to QM, QFT, GR, and relational frameworks.

12.1 Conceptual Implications: Entanglement as Geometry

In standard QM, entanglement is a property of state vectors in a Hilbert space. In the MCM, entanglement is a property of continuum geometry. This shift in perspective has several conceptual consequences.

1. Ontological shift from algebra to geometry. Entanglement becomes a property of the continuum substrate rather than an abstract relation in state space. Non-separability corresponds to geometric overlap of coherence regions:

$$K(x_1, x_2; \Lambda_\Omega) \neq K(x_1; \Lambda_\Omega)K(x_2; \Lambda_\Omega).$$

2. Coherence as a physical field. The coherence field A_C generated by D_{nl} is physically real. Entanglement therefore has spatial extent, geometric structure, and dynamical organization.

3. Emergence of quantum mechanics. Linearity, superposition, and tensor products arise from the spectral properties of continuum operators rather than from fundamental postulates.

4. Nonlocality without action at a distance. Nonlocal correlations arise from the nonlocal operator D_{nl} , which acts on a spatially extended continuum—without the conceptual tension of instantaneous nonlocality in QM.

12.2 Physical Implications: Coherence Regions and Geometric Modulation

The continuum substrate leads to new physical phenomena.

1. Coherence regions. Entanglement is carried by spatial regions Ω_C in which the kernel is approximately constant. These regions can be shaped, stretched, or perturbed by geometric deformation.

2. Geometry-dependent entanglement. Curvature, confinement, and boundary conditions modulate entanglement strength:

$$K(x, y; \Lambda_\Omega) = F(d_g(x, y), \mathcal{J}(x, y; \Lambda_\Omega)).$$

3. Persistent coherence. Because coherence is a field, entanglement can persist longer than QM predicts—even after measurement-like interactions.

4. Deviations from exchange symmetry. Electron vortex modes possess geometric invariants; exchange symmetry arises from topology rather than antisymmetrization. Asymmetric geometries can induce small deviations.

12.3 Technological Implications: Continuum Engineering

If entanglement is geometric, it can be constructed and manipulated.

- 1. Geometry-controlled entanglement devices.** Quantum devices could use curvature, confinement, or boundary conditions to control entanglement.
- 2. Coherence-field amplifiers.** Nonlocal operators D_{nl} suggest mechanisms for amplifying coherence fields, enabling long-range entanglement.
- 3. Continuum-based quantum sensors.** Sensors could detect geometric distortions via changes in coherence fields—a new form of quantum metrology.
- 4. Robust vortex-based qubits.** Electron vortex modes possess topological invariants, suggesting qubits with enhanced stability against decoherence.

12.4 Implications for the Foundations of Quantum Theory

Comparison with QM. QM treats entanglement algebraically; MCM treats it geometrically. QM has no physical mechanism for nonlocality; MCM provides one.

Comparison with QFT. QFT describes nonlocality but without a geometric substrate. MCM provides a continuum that unifies geometry and coherence.

Comparison with GR. GR describes geometry but no quantum nonlocality. MCM proposes a substrate that can carry both curvature and coherence.

Comparison with relational QM. RQM interprets entanglement as relation; MCM provides a physical substrate for these relations.

Comparison with GRW and Everett. GRW collapse becomes geometric disturbance; Everett branching becomes spectral decomposition of continuum operators.

12.5 Philosophical Implications

1. Nonlocality becomes structural. Entanglement is a structural property of the continuum rather than a mysterious algebraic feature.

2. Measurement becomes geometric. Measurement corresponds to geometric disturbance of coherence fields, not collapse of abstract wave functions.

3. Reality becomes monistic. The continuum is the single fundamental entity; particles and quantum states are emergent.

12.6 Result: Conceptual and Technological Impact

Continuum-based entanglement in the MCM implies:

1. Entanglement is geometric, not algebraic.
2. Coherence fields are physically real and manipulable.
3. Quantum mechanics emerges from continuum dynamics.
4. Geometry-dependent entanglement enables new technologies.
5. Foundational interpretations gain a physical substrate.

Thus, the MCM provides both a conceptual unification of quantum and geometric phenomena and a technological perspective for constructing entanglement through continuum geometry.

13 Conclusion

This section addresses the question: What fundamental insights does the MCM provide about the nature of entanglement?

The Monistic Continuum Model (MCM) interprets entanglement as a geometric and dynamical property of a continuous substrate

$$C : M \times \Lambda \rightarrow V,$$

with coherence encoded in the correlation kernel

$$K(x, y; \lambda) = \langle C(x, \lambda), C(y, \lambda) \rangle_V$$

and whose evolution is governed by

$$\frac{\partial}{\partial \lambda} C = D C, \quad D = D_{\text{loc}} + D_{\text{nl}}.$$

This continuum-based perspective provides several fundamental insights into the nature of entanglement, its origin, and its physical significance.

13.1 Entanglement as a Geometric Phenomenon

The MCM shows that entanglement arises from non-separable geometric configurations of the continuum. In contrast to standard QM, where entanglement is an algebraic property of tensor-product Hilbert spaces, the MCM demonstrates that entanglement corresponds to non-factorizable kernel structures:

$$K(x_1, x_2; \lambda) \neq K(x_1; \lambda)K(x_2; \lambda).$$

Entanglement is therefore a structural property of the continuum rather than merely a mathematical relation.

13.2 Coherence Fields as Physical Entities

The coherence field

$$A_C = D_{\text{nl}}(M_1 + M_2)$$

is physically real. It possesses spatial extent, geometric structure, and interacts with vortex modes. Thus, the MCM provides a physical mechanism for nonlocal correlations and replaces the abstract notion of instantaneous nonlocality in QM with a continuous, spatially extended coherence process.

13.3 Emergence of Quantum Mechanics

The MCM shows that quantum mechanics is not fundamental but emergent. Hilbert-space structures arise from the spectral decomposition of the correlation operator

$$(\mathcal{K}_\lambda f)(x) = \int_M K(x, y; \lambda) f(y) d\mu(y),$$

and linear Schrödinger dynamics emerges through coarse-graining of continuum evolution:

$$i \frac{d}{dt} \Psi(t) = \mathcal{H} \Psi(t).$$

Superposition, linearity, and tensor products are therefore effective descriptions of deeper continuum dynamics.

13.4 Unified Geometric Substrate

The continuum substrate provides a unified geometric foundation for quantum and classical phenomena:

- QM emerges from the spectral properties of continuum operators.
- QFT-like excitations appear as vortex modes.
- GR-like geometric effects arise from the manifold structure M .

Quantum nonlocality and classical geometry become manifestations of a single underlying continuum.

13.5 Nonlocality Without Paradox

The MCM resolves conceptual tensions surrounding quantum nonlocality. Nonlocal correlations arise from the nonlocal operator D_{nl} , which acts on a spatially extended continuum—without invoking the notion of instantaneous action at a distance. Entanglement becomes a natural consequence of continuum structure rather than a puzzling quantum effect.

13.6 Technological Implications

The continuum-based view of entanglement opens new technological possibilities:

- geometry-controlled entanglement devices,
- coherence-field amplifiers,
- vortex-based qubits with topological stability,
- continuum-based quantum sensors.

These technologies exploit the geometric nature of entanglement rather than relying solely on algebraic quantum operations.

13.7 Comparison with QM, QFT, and GR

QM. Entanglement is algebraic; the MCM shows it is geometric.

QFT. Nonlocal correlations exist but lack a geometric origin; the MCM provides one.

GR. Geometry exists but without quantum nonlocality; the MCM unifies both.

13.8 Fundamental Insights

The MCM provides the following fundamental insights into entanglement:

1. Entanglement is a geometric property of a continuous substrate.
2. Nonlocal coherence fields are physically real and spatially extended.
3. Quantum mechanics emerges from continuum dynamics.
4. Nonlocality arises from operator structure, not from instantaneous action at a distance.
5. Geometry and quantum coherence are unified within a single framework.

13.9 Final Statement

The MCM shows that entanglement is not an abstract algebraic artifact but a continuum-geometric phenomenon. It provides a physically grounded mechanism for nonlocal correlations, unifies quantum and geometric structures, and opens new conceptual and technological pathways. In doing so, it offers a deeper understanding of entanglement and its role in the fabric of physical reality.

Glossary of Symbols and Concepts

M Smooth manifold representing the geometric substrate of the continuum.

Λ_Ω Organization parameter or internal configuration space controlling continuum evolution.

V Internal vector space encoding local degrees of freedom of the continuum field.

$C(x, \Lambda_\Omega)$ Continuum field; fundamental physical entity of the MCM.

D_α Local operator (stress operator) acting pointwise on C ; preserves separability.

D_{nl} Nonlocal operator generating continuum coherence via an interaction kernel $\mathcal{J}$.

$D = D_{\text{loc}} + D_{\text{nl}}$ Total continuum operator governing evolution in Λ_Ω .

$\mathcal{J}(x, y; \Lambda_\Omega)$ Nonlocal interaction kernel; source of continuum entanglement.

M_1, M_2 Vortex modes; stable localized excitations of the continuum field.

A_C Coherence field generated by D_{nl} ; responsible for nonlocal coupling.

$K(x, y; \Lambda_\Omega)$ Correlation kernel encoding geometric and coherence-based correlations.

$\mathcal{K}_{\Lambda\Omega}$ Correlation operator defined by the kernel K ; admits a spectral decomposition.

ϕ_n Eigenfunctions of $\mathcal{K}_{\Lambda\Omega}$; mixed modes indicate entanglement.

κ_n Eigenvalues of the correlation operator; measure of spectral coherence.

Ω_C Coherence region where the kernel is approximately constant.

W_n Vortex-mode family; geometric building blocks of coupling structures.

$\mathcal{K}_{12}$ Coupling geometry underlying two-electron entanglement.

$\hat{P}_{\text{QM}}$ Spectral projection mapping continuum structures to emergent QM states.

$|\Psi_{12}\rangle$ Emergent two-electron quantum state arising from continuum geometry.

14 References

This section provides the sources relevant to the scientific questions addressed throughout this work.

The Monistic Continuum Model (MCM) integrates concepts from quantum mechanics (QM), quantum field theory (QFT), general relativity (GR), geometric and topological field theory, nonlocal operator theory, and relational quantum frameworks. The following references form the scientific foundation for the mathematical structures, operator formalisms, functional spaces, dynamical equations, and entanglement concepts discussed in this work. All references are formatted in IOP-compliant style.

14.1 Quantum Mechanics and Entanglement Theory

- J. von Neumann, *Mathematical Foundations of Quantum Mechanics*, Princeton University Press (1955).
- A. Einstein, B. Podolsky, N. Rosen, “Can Quantum-Mechanical Description of Physical Reality Be Considered Complete?”, *Phys. Rev.* **47**, 777 (1935).
- J. S. Bell, “On the Einstein Podolsky Rosen Paradox”, *Physics* **1**, 195 (1964).
- M. A. Nielsen, I. L. Chuang, *Quantum Computation and Quantum Information*, Cambridge University Press (2000).
- R. Horodecki et al., “Quantum entanglement”, *Rev. Mod. Phys.* **81**, 865 (2009).

14.2 Quantum Field Theory and Nonlocal Correlations

- S. Weinberg, *The Quantum Theory of Fields*, Vols. I–III, Cambridge University Press (1995).
- M. E. Peskin, D. V. Schroeder, *An Introduction to Quantum Field Theory*, Addison-Wesley (1995).
- R. Haag, *Local Quantum Physics*, Springer (1996).
- H. Halvorson, M. Müger, “Algebraic Quantum Field Theory”, in *Handbook of Philosophy of Physics*, Elsevier (2007).

14.3 General Relativity, Geometry, and Continuum Theory

- C. W. Misner, K. S. Thorne, J. A. Wheeler, *Gravitation*, W. H. Freeman (1973).
- S. W. Hawking, G. F. R. Ellis, *The Large Scale Structure of Space-Time*, Cambridge University Press (1973).
- R. M. Wald, *General Relativity*, University of Chicago Press (1984).
- M. Spivak, *A Comprehensive Introduction to Differential Geometry*, Publish or Perish (1979).

14.4 Geometric and Topological Field Theory

- M. Nakahara, *Geometry, Topology and Physics*, 2nd ed., Institute of Physics Publishing / CRC Press (2003).

14.5 Operator Theory, Functional Analysis, and Kernel Methods

- M. Reed, B. Simon, *Methods of Modern Mathematical Physics*, Vols. I–IV, Academic Press (1972–1978).
- T. Kato, *Perturbation Theory for Linear Operators*, Springer (1995).
- W. Rudin, *Functional Analysis*, McGraw-Hill (1991).
- I. Daubechies, *Ten Lectures on Wavelets*, SIAM (1992).

14.6 Collapse Models (GRW) and Alternative Quantum Frameworks

- G. C. Ghirardi, A. Rimini, T. Weber, “Unified dynamics for microscopic and macroscopic systems”, *Phys. Rev. D* **34**, 470 (1986).
- P. Pearle, “Combining stochastic dynamical state-vector reduction with spontaneous localization”, *Phys. Rev. A* **39**, 2277 (1989).
- A. Bassi, G. C. Ghirardi, “Dynamical reduction models”, *Phys. Rep.* **379**, 257 (2003).

14.7 Everett, Relational QM, and Interpretative Frameworks

- H. Everett, “Relative State Formulation of Quantum Mechanics”, *Rev. Mod. Phys.* **29**, 454 (1957).
- D. Wallace, *The Emergent Multiverse*, Oxford University Press (2012).
- C. Rovelli, “Relational Quantum Mechanics”, *Int. J. Theor. Phys.* **35**, 1637 (1996).
- C. Brukner, “On the quantum measurement problem”, in *Quantum [Un]Speakables II*, Springer (2017).

14.8 Nonlocality, Coherence, and Kernel-Based Physics

- A. Zeilinger, “Experiment and the foundations of quantum physics”, *Rev. Mod. Phys.* **71**, S288 (1999).
- N. Brunner et al., “Bell nonlocality”, *Rev. Mod. Phys.* **86**, 419 (2014).
- S. Popescu, D. Rohrlich, “Quantum nonlocality as an axiom”, *Found. Phys.* **24**, 379 (1994).

14.9 Mathematical Foundations Relevant to the MCM

- L. Hörmander, *The Analysis of Linear Partial Differential Operators*, Springer (1983).
- J. M. Lee, *Introduction to Smooth Manifolds*, Springer (2012).
- S. Lang, *Real and Functional Analysis*, Springer (1993).
- M. Atiyah, “Topological quantum field theories”, *Publ. Math. IHÉS* **68**, 175 (1988).

14.10 Summary

These references collectively support the mathematical, physical, and conceptual foundations of the MCM, including:

1. continuum geometry and manifold theory,
2. operator theory and functional analysis,
3. quantum entanglement and nonlocality,
4. vortex dynamics and topological field structures,
5. emergent quantum mechanics,
6. comparisons with QM, QFT, GRW, Everett, and relational QM.

They provide the scientific basis for the continuum-geometric interpretation of entanglement developed throughout this work.

15 Citation Recommendation

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15.2 BibT_EX

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